CHAPTER 2 • GRADE 4 • WEEK 1

Moving Objects Have Energy

MOTION • CHANGE • EVIDENCE

How are energy, change, and movement of objects related?

MOVING OBJECT → CHANGE → EVIDENCE



Unit 1 • Energy Transfer & Transformation

Big Question

BIG QUESTION

How are energy, change, and movement of objects related?

Keep this question in mind. We will return to it.

Think: What is ONE change a moving object can cause?

Big Question: My first thinking

Guided Notes p. 1

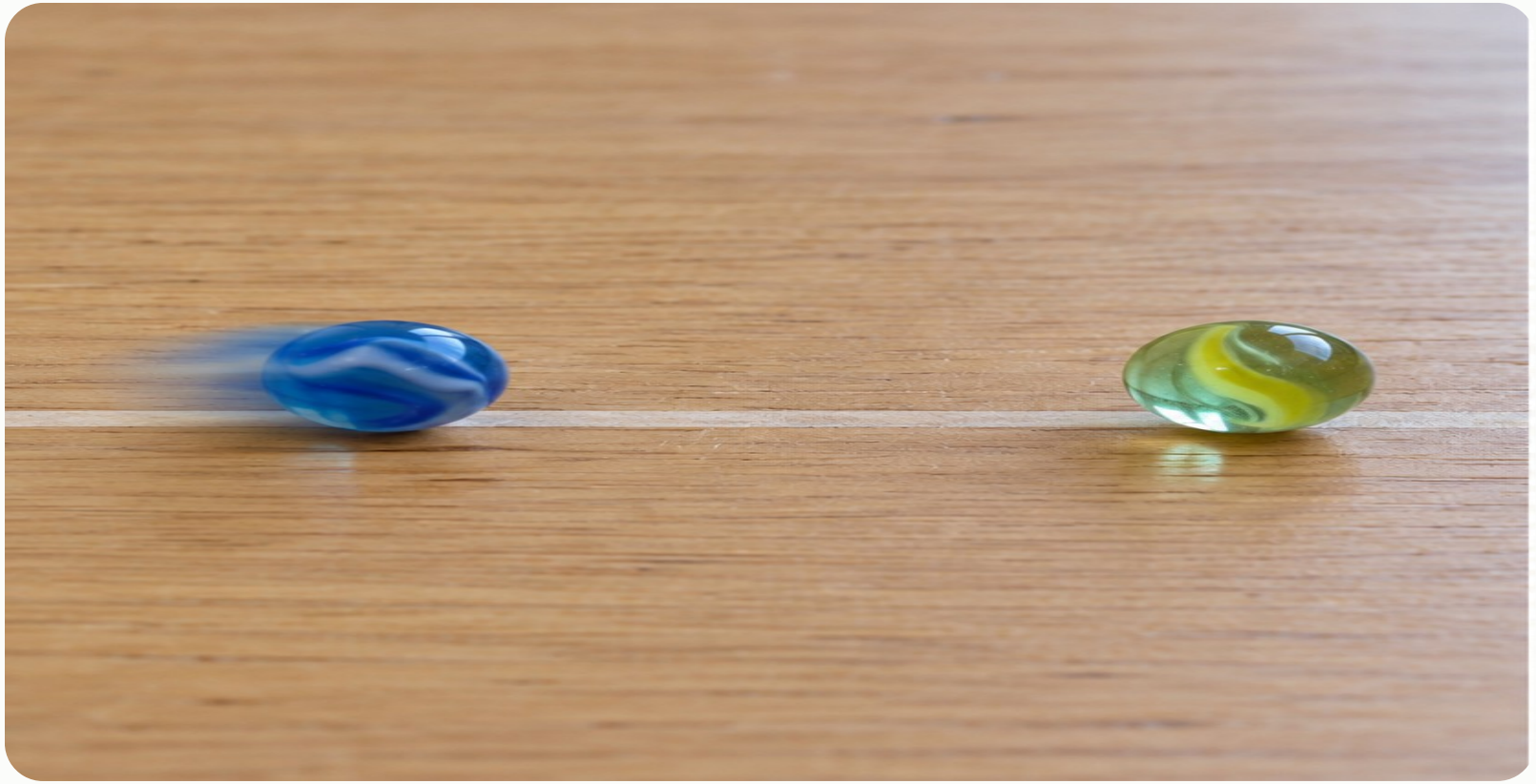
How are energy, change, and movement of objects related?

My first thinking

Write 1–2 quick ideas. No wrong answer yet.

Marble Collision: Before

One marble is still. Another marble will roll toward it.



BEFORE

One marble is still.

One marble is ready to move.

What do you notice before the marbles touch?

Predict: What will happen?

Think before the moving marble reaches the still marble.

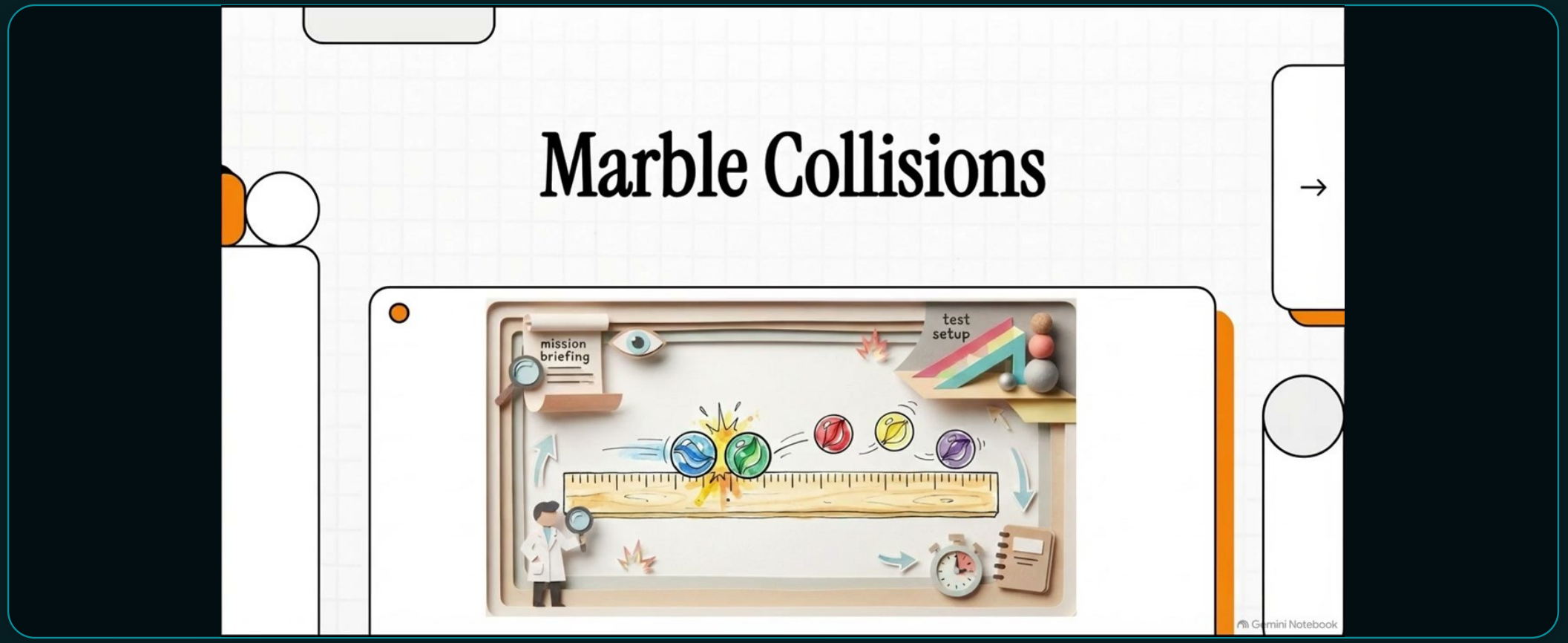
When the moving marble hits the still marble, the still marble will...



Say your prediction to a partner.

OPTIONAL SUPPORT

Marble Collision Replay



What changes after the marbles touch?

Live demonstration is the default. Continue to Slide 7 if video is unavailable.

What changed?



CHANGE

The target marble changed from still to moving.

Describe only what you observed.

How do we know it had energy?

Energy can be invisible. We look for the change.

MOVING OBJECT → **CHANGE** → **EVIDENCE**

The target marble started to move. That change is evidence that the moving marble had energy.

We observe the change. We infer that energy was involved.

Vocabulary

KEY WORD

motion

the process of an object changing position



Reader p. 7

Vocabulary: Motion

Guided Notes • Reader p. 7

motion – the process of an object changing _____ .

Write the missing word.

Vocabulary: Motion

Guided Notes • Reader p. 7

motion – the process of an object changing position .

position

How did each object begin to move?

Reader p. 8





What started the motion in each example?

Write the Missing Words

Guided Notes • Reader p. 8

A moving object can cause a _____ .

Players cause their own motion. A ball does not. Something else puts the ball into motion.

Some objects cause their own motion. Other objects are put into motion by _____ motion.

Fill both blanks in order.

Moving Objects Can Cause Changes

Guided Notes • Reader p. 8

A moving object can cause a change .

Players cause their own motion. A ball does not. Something else puts the ball into motion.

**Some objects cause their own motion. Other objects are put into motion by
outside motion.**

change • outside

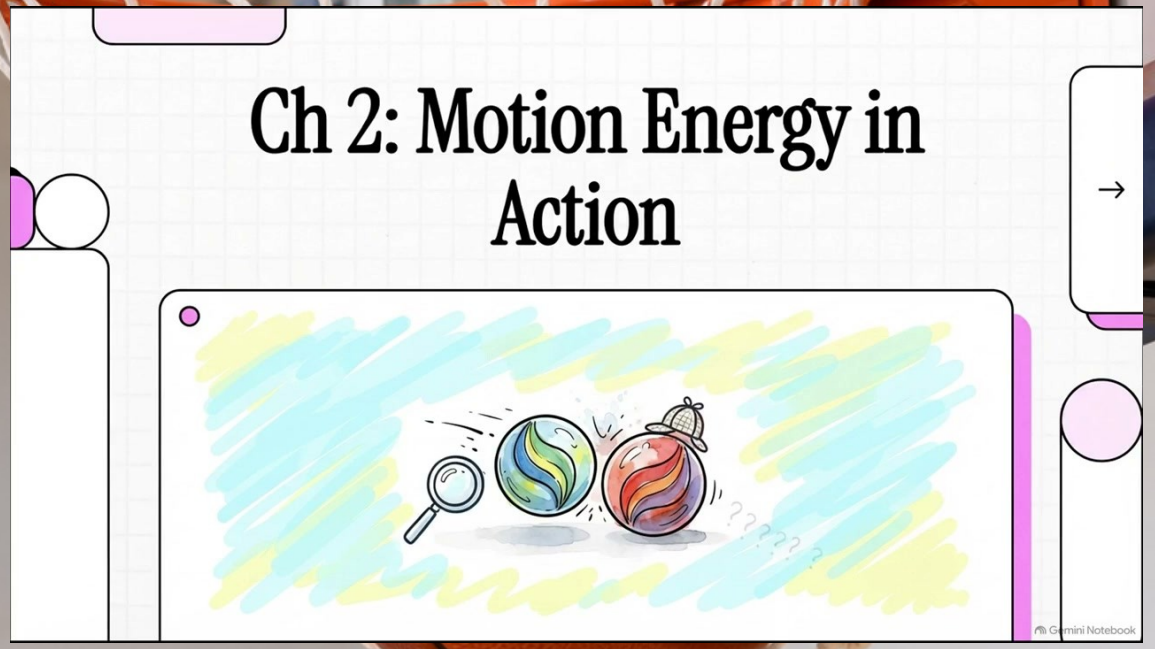
R • A OBSERVE

Look for the change

Reader pp. 8–9

What changed because the ball was moving?

MOVING BALL → **NET CHANGES**



Vocabulary

KEY WORD

energy of motion

the energy an object possesses while it is moving



Reader p. 9

Vocabulary: Energy of Motion

Guided Notes • Reader p. 9

energy of motion – the energy an object possesses while it is _____ .

Write the missing word.

Vocabulary: Energy of Motion

Guided Notes • Reader p. 9

energy of motion – the energy an object possesses while it is moving .

moving

Moving objects have energy

Reader p. 9



Energy is the ability to cause change.

A moving object causes change, so it has energy.

Trace the model from left to right.

Moving Objects Have Energy

Guided Notes • Reader p. 9

All moving objects have energy of _____ .

Energy is the ability to cause change. A moving object causes change, so it has energy.

MOVING OBJECT → **CHANGE** → **EVIDENCE**

Write the missing word.

Marble Collision: Observe

Guided Notes • Reader p. 9

OBSERVE: one moving marble rolls into a target marble.

BEFORE

draw

→

AFTER

draw

What changed?

Moving Objects Have Energy

Guided Notes • Reader p. 9

All moving objects have energy of motion .

MOVING OBJECT → **CHANGE** → **EVIDENCE**

Sample observation

The target marble changed from sitting still to moving.

The change matters more than the drawing style.

People use the energy of motion

Reader pp. 10–12



moving hammer

What changes?



swinging pendulum

What moves?



moving air

What changes?

People Use the Energy of Motion

Guided Notes • Reader pp. 10–12

Write or draw the change each moving object causes.

MOVING OBJECT	WHAT CHANGE DOES IT CAUSE?
hammer (swinging)	
pendulum (swinging)	
wind (moving air)	

People Use the Energy of Motion

Guided Notes • Reader pp. 10–12

SAMPLE ANSWERS – YOUR WORDING MAY DIFFER

MOVING OBJECT	WHAT CHANGE DOES IT CAUSE?
hammer (swinging)	drives a nail into the wood
pendulum (swinging)	turns the clock gears
wind (moving air)	spins the turbine blades

Check Your Thinking

Guided Notes • Reader pp. 7–12

How do we know a moving object has energy?

Start with: "We know because the moving object causes …"

Check Your Thinking

Guided Notes • Reader pp. 7–12

SAMPLE RESPONSE – WORDING MAY VARY

“We know because the moving object causes a change, like a moving marble making another marble move.”



A VOTE

Which example shows energy of motion?

A moving marble makes a still marble move.

A rock keeps sitting still on the ground.

A pencil rests on a desk.

Look for a moving object that causes a change.

Observation and inference

OBSERVATION

The target marble moved.

What we directly saw

INFERENCE

The moving marble had energy.

What the change tells us

Big Question: Now I think

Guided Notes p. 3

How are energy, change, and movement of objects related?

Now I think...

Use what you learned about motion and energy of motion.

Chapter 2 Check

Moving Objects Have Energy

12 POINTS

1. **Write your name and homeroom.**
2. **Read each question carefully.**
3. **Use complete thoughts when a question asks you to explain.**
4. **When you finish, turn your paper over and wait quietly.**

You will have about 12 minutes. Teacher chooses open note or closed note.

Review 1: concept

A moving object has energy. What is this energy called?

A stored energy

B energy of motion

C energy of light

D energy of sound

Review 1: Answer

A moving object has energy. What is this energy called?

A stored energy

✓ B **energy of motion**

C energy of light

D energy of sound

Energy of motion (Reader p. 9).

Review 2: visual

A marble rolls and hits a still marble. What change do you observe?

A. Both marbles stop moving

B. The still marble starts to move

C. The marbles change color

D. Nothing happens

Review 2: Answer

A marble rolls and hits a still marble. What change do you observe?

A Both marbles stop moving

✓ B **The still marble starts to move**

C The marbles change color

D Nothing happens

The still marble starts to move (Reader p. 8).

Review 3: scenario

A swinging hammer hits a nail. What change does the hammer cause?

A. The hammer turns to liquid

B. The hammer floats away

C. The nail is driven into the wood

D. The nail melts

Review 3: Answer

A swinging hammer hits a nail. What change does the hammer cause?

A The hammer turns to liquid

B The hammer floats away

✓ C **The nail is driven into the wood**

D The nail melts

The hammer drives the nail in (Reader p. 10).

Review 4: application

How do we know a moving object has energy if we cannot see the energy itself?

A We can always see energy directly

B We guess without any evidence

C Moving objects never have energy

D We observe the change it causes

Review 4: Answer

How do we know a moving object has energy if we cannot see the energy itself?

A We can always see energy directly

B We guess without any evidence

C Moving objects never have energy

✓ D **We observe the change it causes**

We know because it causes a change.

Review 5: scenario

Wind (moving air) spins windmill blades. What does this show?

A Moving air has energy of motion

B Still air is stronger

C Windmills work without energy

D Blades move on their own

Review 5: Answer

Wind (moving air) spins windmill blades. What does this show?

✓ A **Moving air has energy of motion**

B Still air is stronger

C Windmills work without energy

D Blades move on their own

Moving air has energy of motion (Reader p. 12).

Moving objects have energy because they can cause change.

Moving Marbles

OPTIONAL • NOT GRADED

Can a moving marble cause an observable change?



MOVING MARBLE → CHANGE → EVIDENCE

Materials and safety

OPTIONAL • NOT GRADED



Materials

- two identical marbles
- clear testing lane
- Moving Marbles sheet
- pencil

Roll, do not throw.
Keep hands out of the lane during a roll.
Retrieve marbles after the trial stops.

Before you roll

OPTIONAL • NOT GRADED

What do you think will change when the moving marble reaches the target marble?



Record your prediction on the Moving Marbles sheet.

Run one fair observation trial

OPTIONAL • NOT GRADED

1. Place the target marble in the marked spot.
2. Roll the moving marble toward it.
3. Watch what moves before contact.
4. Watch what changes after contact.
5. Record only what you observed.

Keep the setup the same. Do not measure speed.

Show what happened

OPTIONAL • NOT GRADED

BEFORE



AFTER

What changed?

Evidence discussion

OPTIONAL • NOT GRADED

What change is evidence that the moving marble had energy?

I know the moving marble had energy because _____.

MOVING OBJECT → CHANGE → EVIDENCE

One possible observation

OPTIONAL • NOT GRADED



SAMPLE – YOUR TRIAL MAY LOOK DIFFERENT

The target marble was still before the collision. After the collision, it moved.

That change is evidence that the moving marble had energy.

Chapter 2 close

OPTIONAL • NOT GRADED

Moving objects have energy because they can cause change.

If the SAME object moves faster, could it cause a bigger change?

NEXT: CHAPTER 3 • ENERGY AND SPEED ARE RELATED